

Student Name	MONISHA S
Register Number	TU6253210211034
Project Title	Multi-Echelon Supply Chain Network Design via Mixed-Integer Linear Programming (MILP)
Week	Week 3

WEEK-3 Technical Report

Task Name

Multi-Echelon Supply Chain Network Design via Mixed-Integer Linear Programming (MILP)

Abstract

This project focuses on designing a supply chain network involving factories, warehouses, and customer zones. The aim is to minimize transportation and storage costs while satisfying customer demand using Mixed-Integer Linear Programming (MILP).

Introduction

A multi-echelon supply chain consists of multiple stages through which products move from factories to customers. Optimizing these stages helps reduce costs and improve overall efficiency.

Objectives

- Minimize transportation and storage costs.
- Satisfy customer demand.
- Optimize product movement across the supply chain.
- Improve operational efficiency.

Supply Chain Components

Factories

Manufacture products and supply them to warehouses.

Warehouses

Store products and distribute them to customers.

Customer Zones

Receive products based on demand requirements.

MILP Model

Objective Function

Minimize the total operational cost of the supply chain.

Constraints

- Factory production capacity.
- Warehouse storage capacity.
- Customer demand fulfillment.
- Product flow balance.

Methodology

- Define factories, warehouses, and customer zones.
- Collect supply, demand, and cost data.
- Formulate the MILP model.
- Apply capacity and demand constraints.
- Solve the optimization problem.

Simple Python Code

```
from scipy.optimize import linprog
```

```
c = [4, 6, 5, 3]
```

```
A = [[1, 1, 0, 0],  
      [0, 0, 1, 1]]
```

```
b = [100, 120]

result = linprog(c, A_ub=A, b_ub=b, bounds=(0, None))
```

```
print("Minimum Cost:", result.fun)
```

Results

The optimized network reduces transportation costs, efficiently utilizes warehouse capacity, and satisfies customer demand.

Applications

- Manufacturing industries
- Retail supply chains
- E-commerce logistics
- Distribution networks

Conclusion

MILP provides an effective way to optimize supply chain operations by reducing costs and improving product distribution across multiple stages of the network.